

Dou Li

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Links: [Google Scholar](#) (h-index: 6) | [ResearchGate](#) | [ORCID](#)

Research Interests

- Coupled physical-biogeochemical processes in estuarine-shelf ecosystems (e.g., eutrophication, hypoxia)
- Mesoscale and submesoscale dynamics and their biogeochemical impacts in coastal upwelling systems
- Ocean iron fertilization and marine carbon dioxide removal

Employment

University of California, Los Angeles, Department of Atmospheric and Oceanic Sciences, Los Angeles, CA, USA Mar 2025 – present
Postdoctoral Scholar, Advisor: Prof. Daniele Bianchi

Boston University, Department of Earth & Environment, Boston, MA, USA Jul 2022 – Feb 2025
Postdoctoral Associate, Advisor: Prof. Xiaozhou Ruan

Education

The Hong Kong University of Science and Technology (HKUST) Aug 2017 – Jun 2022
Ph.D. in Marine Environmental Science, Advisor: Prof. Jianping Gan
Thesis: Biophysical Dynamics on the Hypoxia Formation and Variability in the Pearl River Estuary: Observational and Modeling Studies

Sun Yat-sen University (SYSU), China Sep 2013 – Jul 2017
B.S. in Marine Science

Peer-Reviewed Publications

1. Buesseler, K., Bianchi, D., Burns, W., Chai, F., Cullen, J., Estapa, M., Honda, M., John, S., Karl, D., Klaas, C., **Li, D.**, Liu, Z., McGillicuddy, D., Morris, P., Nishioka, J., Omand, M., Ramakrishna, K., Roche, K., Siegel, D., Smith, S., Twining, B., Wells, M., White, A., Xiu, P., Yoon, J. (2026). The Case for Ocean Iron Fertilization Field Trials. *Dialogues on Climate Change* (accepted)
2. Luo, B., Zhang, S., Takeuchi, N., Lumley, D., Beck, S., **Li, D.** and Zhu, H.* (2026). Deep seafloor seismic monitoring for oceanic and atmospheric pressure changes. *Journal of Geophysical Research-Solid Earth* (under review, IF: 4.1).
3. **Li, D.**, Ruan, X.*, Pham, A., Damien, P. and Bianchi, D. (2025). The role of eddies in primary production off the California Current System. *Geophysical Research Letters*, 52(19), e2025GL118614. <https://doi.org/10.1029/2025GL118614> (IF: 4.6)
4. **Li, D.**, Wei, H., Ruan, X.* (2025). The importance of eddy stirring in wind-driven coastal upwelling. *Journal of Physical Oceanography*, 55(1), 29-42. <https://doi.org/10.1175/JPO-D-24-0066.1> (IF: 3.0)
5. Yu, L., Gan, J.*, **Li, D.**, Cheng, W., Zhang, Y., Kung, H., Hui, C., & Chen, Z. (2025). Shellfish Aquaculture Complements Nutrient Reduction in Mitigating Climate-Exacerbated Coastal Hypoxia. *Environmental Science & Technology*, 59(35), 18687-18695. <https://doi.org/10.1021/acs.est.5c06682> (IF: 11.3)
6. **Li, D.***, & Ruan, X. (2024). On the Pathways of Wind-Driven Coastal Upwelling: Nonlinear Momentum Flux and Baroclinic Instability. *Journal of Physical Oceanography*, 54(1), 63-79 <https://doi.org/10.1175/JPO-D-23-0098.1> (IF: 3.0)
7. **Li, D.**, Gan, J.*, Lu, Z., Cheng, W., Kung, H., & Li, J. (2023). Hypoxia formation triggered by the organic matter from subsurface chlorophyll maximum in a large estuary-shelf system. *Water Research*, 120063. <https://doi.org/10.1016/j.watres.2023.120063> (IF: 12.4)
8. **Li, D.**, Gan, J.*, Hui, C., Yu, L., Liu, Z., Lu, Z., et al. (2021). Spatiotemporal development and dissipation of hypoxia induced by variable wind-driven shelf circulation off the Pearl River Estuary: Observational and

modeling studies. *Journal of Geophysical Research: Oceans*, 126. <https://doi.org/10.1029/2020JC016700> (IF: 3.4)

9. **Li, D.**, Gan, J.*, Hui, R., Liu, Z., Yu, L., Lu, Z., & Dai, M. (2020). Vortex and biogeochemical dynamics for the hypoxia formation within the coastal transition zone off the Pearl River Estuary. *Journal of Geophysical Research: Oceans*, 125. <https://doi.org/10.1029/2020JC016178> (Issue Cover). (IF: 3.4)
10. Yu, L., Gan, J.*, Dai, M., Hui, C. R., Lu, Z., & **Li, D.** (2021). Modeling the role of riverine organic matter in hypoxia formation within the coastal transition zone off the Pearl River Estuary. *Limnology and Oceanography*, 66(2), 452-468. <https://doi.org/10.1002/lno.11616> (2020-2021 Top Cited Article in Limnology and Oceanography, IF: 3.7)
11. Cheng, M., Liu, M., **Li, D.**, Luo, Q., Zhang, Z., Yuan, L., ... & Wang, X. (2021). Human methylmercury exposure and potential impacts in central Tibet: food and traditional Tibetan medicine. *Bulletin of Environmental Contamination and Toxicology*, 1-10. <https://doi.org/10.1007/s00128-021-03216-5> (IF: 2.6)

Books

1. The First Scientific Assessment Report on Oceans and Climate Change, Ocean Publications, ISBN 978-7-5210-0571-4, Member of Editing and Production Committee (2019). (On the list of excellent marine science and technology books in 2021).

Manuscripts in preparation

1. **Li, D.***, Damien, P. Guiet, J., and Bianchi, D. The Role of Mesoscale and Submesoscale Processes in Ocean Iron Fertilization in the Northeast Pacific Ocean.
2. **Li, D.** & Gan, J. Heterogenetic submesoscale plume front dynamics in an estuary-shelf system.

Conference Presentations and Invited Talks

CalGFD , UCSB, Santa Barbara, CA, USA	Sep 2025
Oral presentation: "Role of eddies in primary production in the California Current System"	
Marine Center Spring Meeting (invited) , UCLA, Los Angeles, CA, USA	Jun 2025
Oral presentation: "Role of eddies in primary production in the California Current System"	
University of California-Los Angeles (invited) , Los Angeles, CA, USA	Mar 2024
Oral presentation: "Unveiling three-dimensional nutrient pathways in wind-driven coastal upwelling"	
Ocean Sciences Meeting (OSM) 2024 , New Orleans, LA, USA	Feb 2024
Oral presentation: "The importance of eddy stirring in wind-driven coastal upwelling"	
12th International Workshop on Modeling the Ocean (IWMO) , Ann Arbor, Michigan, USA (online)	Jul 2022
Oral presentation: "Heterogeneous submesoscale plume front dynamics in an estuary-shelf system"	
Symposium on OCEAN_HK 2021 , Hong Kong, China	Jul 2021
Oral presentation: "Submesoscale processes in plume-affected coastal ocean"	
Symposium on OCEAN_HK 2020 , Hong Kong, China	Aug 2020
Oral presentation: "Physical control and biogeochemical response of variable hypoxia in the coastal transition zone off the Pearl River Estuary"	
Annual conference of Qingdao Marine Science and Technology Pilot National Laboratory , Qingdao, China	Jan 2020
Oral presentation: "Vortex and biogeochemical dynamics for hypoxia formation over a coastal transition zone"	
Symposium on OCEAN_HK 2019 , Hong Kong, China	Jun 2019
Oral presentation: "Variable biophysical characteristics and hypoxia during 2017-18 cruises"	
Annual meeting of CHOICE-C II Carbon Cycle in South China Sea: budget, controls & global implications , Tianjin, China	May 2019
Oral presentation: "Physical characteristics of 2017-18 OCEAN-HK Cruise"	
4th Xiamen Symposium on Marine Environmental Science (XMAS) , Xiamen,	Jan 2019

China

Poster presentation: "Variability of hypoxia in coastal transition zone off Pearl River Estuary: observational and modeling studies"

Research Experience

Exploring Ocean Iron Solutions (ExOIS) Project: <i>Multiscale observing system simulation experiments for iron fertilization in the Southern Ocean, Equatorial Pacific, and Northeast Pacific, (PI: Dennis McGillicuddy)</i> Foundation: NOAA Global Ocean Monitoring and Observing (GOMO), NOAA Ocean Acidification Program (OAP), National Science Foundation (NSF) Contribution: model development, data analysis, and publication writing	2025–
U.S. National Science Foundation (OCE-2343008): <i>Toward a more complete understanding of coastal upwelling dynamics, (PI: Xiaozhou Ruan)</i> Contribution: model implementation, data analysis, and publication writing	2024–2025
National Natural Science Foundation of China (Key Program): <i>On the internal dynamics of layered circulation in the South China Sea, (PI: Jianping Gan)</i> Contribution: Co-authored and contributed to proposal writing	2020–2024
Hong Kong Research Grant Council Theme-based Research Scheme: <i>Diagnosis and prognosis of intensifying eutrophication, hypoxia, and ecosystem consequences around Hong Kong waters: coupled physical–biogeochemical–pollution studies, (PI: Jianping Gan)</i> Contribution: model development, data analysis, and publication writing	2017–2022

Teaching and Mentorship

Graduate Student Mentor at University of California, Los Angeles	2025–
Undergraduate Student Mentor at University of California, Los Angeles	2025–
Undergraduate Student Mentor at Boston University	2023–2024
Teaching Assistant in Introduction to Oceanography (OCES5001) at HKUST	2018–2020

Fieldwork and Cruises

Haike68 Research Cruise at Pearl River and Northern South China Sea for 28 days	Jun 2021
Haike68 Research Cruise at Pearl River and Northern South China Sea for 30 days	Jul 2018

Awards and Scholarships

Postdoc Travel Award, Boston University	Feb 2024
Outstanding Young Scientist Award in 12th International Workshop Modeling the Ocean, IWMO	Jun 2022
RedBird Academic Excellence Award, HKUST	Nov 2021
Best Presentation Award, HKUST Postgraduate Seminar	2019/2020
Postgraduate Studentship of HKUST	2017–2022
Excellent Thesis/Outstanding Graduate Student of SYSU	2017
National Scholarship, Ministry of Education of China	2015
First-class Scholarship of SYSU	2015

Second-class Scholarship of SYSU

2014/2016

The Sun Hung Kai Properties' Scholarship for Outstanding Students of SYSU

2014/2015

Additional Information

Journal reviewer: *Nature Communications, Geophysical Research Letter, Journal of Geophysical Research-oceans, Global Change Biology, Science of the Total Environment, Ocean Science, Marine Pollution Bulletin*

Student Presentation Evaluation Program (SPEP) reviewer: *Ocean Science Meeting 2024*

Event organizer: *Southern California Biogeochemical Ocean Observations and Models (SoCal BOOM) 2025*

Language skills: *Mandarin (Native), English*

Numerical modeling: *Regional Ocean Modeling System (ROMS)*

Computer programming: *Fortran, MATLAB, Python*

Field observations skills: *Operation of CTD, ADCP, Turbulence Profiler*